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A Context-Aware Analytical Framework for Quantifying the Influence of Agile Maturity on Enterprise Software Project Success

Abstract—Agile practices are commonly used in the enterprise software development, however degree of how Agile maturity impacts on project success has not been quantitatively measured well; especially with reference to complex enterprise environment. Current research is primarily based on qualitative evaluations, or maturity models specific to a particular framework, which do not provide empirical evidence for how different levels of maturity relate to tangible project results. To fill this void, we present in this paper a Context-Aware Agile Success Estimator (CASE)—a new interpretable analytic model intended to quantitatively appraise the influence of agile maturity on software project success and by explicitly incorporating organizationally specific contextual constraints. The approach considers Agile maturity as a multifaceted measure, extracted from major organizational and process dimensions such as teams' autonomy, the ability for continuous delivery, leadership commitment for delivering added-value to users, the integration of customer feedback into potential workflows and accumulated discipline in applying a given process. Enterprise context is defined by features like organizational size, regulatory burden, team dispersion and complexity of legacy system. Through a weighted linear estimation mechanism, CASE relates these dimensions to multi-criteria project success measures including delivery timeliness, product quality, budget-achievement performance and client satisfaction as well as realized business value. Using a dataset of an agile project from kaggle2024, the framework was empirically confirmed. The results of the experiments show projects on higher levels of agile maturity had better performance on-time 21.6%, post-release defects 17.9% and stakeholder satisfaction 24.3% than low Agile maturity projects. The overall predictive accuracy for project success was 88.7%, demonstrating that the model is robust and have practical implications. Through the provision of a transparent, context-adaptable and quantitatively-founded evaluation methodology this work provides an instrumental decision-support artefact to both small- and large-scale enterprises wanting to anchor their agile transformation endeavours towards concrete project success results.

Keywords—Agile Maturity, Software Project Success, Enterprise Software Engineering, Context-Aware Analytics, Agile Performance Evaluation, Project Outcome Prediction

I Introduction

Enterprise software developers increasingly use agile development methodologies for their superior focus on flexibility, quick feedback loops, and incremental value delivery. The use of Agile practices to convert with uncertainty and enhance

cross-functional cooperation have become ever more popular in complex software projects as organizations strive for faster time-to-market. On the other hand, big organizations are typically bound by regulatory demands and competing management priorities, distributed teams and governance structures of hierarchy, not to mention legacy systems that heavily impact how Agile practices can truly work. Therefore, the results of Agile adoption are quite different from one organization to another, as it is noticeable that an objective understanding was still missing about the factors of success in applying Agile practices in enterprise contexts. One of the most important factors in effective Agile is the amount an organisation has to mature within their Agile adoption. Agile maturity refers to how well Agile principles are institutionalized in teams, processes, leadership tiers and delivery methods. Although several organisations profess to adopt Agile practices, variations in maturity levels result in significantly different project outcomes such as delivery predictability, product quality, cost compliance and stakeholder satisfaction. The existing risk-guided and success-guided approaches for assessing the maturity of Agile-oriented organizations in large-scale enterprises are often qualitative, framework specific or focus on isolated dimensions that hardly support quantifying how maturity levels in turn would impact on measurable project results.

To this end, this research presents a CASE, a quantitative and interpretable analytical model for quantifying how much organisational context influence in which enterprise software projects' success is achieved by Agile maturity. The proposed approach includes modelling Agile maturity as a compound index of certain organisational and process dimensions and incorporates enterprise specific situational constraints in the evaluation. This study seeks to establish the connection between maturity levels and multi-criteria project success indicators using a well-structured forecasting approach; it has the potential to create a practical decision-supporting tool for companies to properly match their Agile transformation initiatives with real-world applicable and quantifiable project success results. This research has three main contributions to the practice and academia in enterprise software engineering. First, we proposed a new CASE that quantitatively characterizes the relationship between process maturity and successful software projects, while capturing the explicit role of enterprise-specific contextual constraints. Second, the proposed approach makes Agile maturity tangible by means of synthesizing it as a composite, interpretable index which considers both organisational and process perspectives and can be used to assess systematically and replicable in heterogeneous enterprise settings. Third, we have conducted an empirical study to show that higher Agile maturity is significantly related with better levels of delivery performance in the dataset from an open source Agile project, better product quality and stakeholder satisfaction (which are all three major characteristics of

a given process), thus establishing the connection between notions of conceptual models and actual outcomes.

1.1 Objectives

- To establish a numeric and interpretable measuring system for Agile maturity and its relationship with the success of enterprise software project.
- To include organizational context and environmental factors in assessing Agile project results for better assessment precision.
- To empirically confirm the proposed framework with real-life Agile project data and multi-criteria success measures.

II Literature Review

Junior et al. (2024) identified critical success factors (CSFs) affecting successful agile software development in a study that 25 factors were categorized by project, people, organizational, process and culture. Based on survey data and structural equation modeling, the research showed that all CSF categories have a significant effect on agile project success with projects and people being the most influential. The results highlight the interaction effect of CSFs when considered as a collective rather than individual factors [1]. Russo (2021) studied an extensive industry based case of large scale agile transformation in a mission-critical context, highlighting the importance of stakeholder commitment as key to success. An agile success model for hierarchical and plan-driven organizations was developed through post-mortem analysis and empirical validation using PLS-SEM. The findings emphasize the need for organisational levels and management acknowledgment to support agile success over long lasting transformation [2]. Khoza and Marnewick (2023) developed a theoretical model for measuring success in scaled agile, focusing on processes and perceived success factors at portfolio, program, and team levels. A correlation and regression analysis performed during the study helped to extract several success drivers at each of the three levels (e.g., strategic investment funding, ongoing exploration). The model helps in understanding the impact of scale agile on organizational success factors [3]. Rashid et al. (2021) proposed a Green-Agile Maturity Model (GAMM) for global software development in the context of sustainability. The model was built from a systematic literature review, expert validation and case studies to categorise practices in seven maturity levels. The study shows how mixing the green and agile elements can improve level of software sustainability and organisation maturity [4]. Moraga-Díaz et al. (2023) studied enterprise agility in a case of agile transformation from a global scale manufacturing company. The analysis established the validity of agility metrics via parametric statistical tests and showed that focused measurement, along with continual

improvement, are key to developing enterprise agility. The work emphasizes the need for quantitative validation to evaluate agility at scale [5].

Fagarasan et al. (2023) presented a data-driven score model and the inclusion of the sustainability metrics in the performance evaluation of agile projects. The model is applied to a case study and it bridges a feature delivery predictability with sustainability goals, which constitutes a gap in agile performance metrics. This is consistent with the requirement for harmonized measures to enable long-term efficiency and sustainable software development [6]. Panjaitan and Legowo (2022) assessed the maturity of Scrum application with a customised Scrum Maturity Model based on the Scrum Guide and Agile Maturity Model. Focus group discussions showed Scrum practices were at the early maturity stage, indicating the misfit between expected and practiced. The study offers practical suggestions to enhance Scrum adoption and process discipline [7]. Garomssa et al. (2022) studied the impact of intellectual capital on the performance of commercial open-source software firms in terms of the mediating role played by software quality. Empirical investigation based on survey data suggests that relational and structural capital influence success indirectly through human capital and the quality dimension. These findings further support that the software quality is vital for sustainable organizational success [8]. The interactive effects of lean construction and BIM on project success was studied by Polat and Demirkesen (2025). Lean practices have a greater direct impact on project success than BIM and they improve the effectiveness of BIM. While performed in the construction field, results suggest similar maturity–success relationships for agile and hybrid project settings [9]. Ansari et al. (2024) examined the impact of transformational and transactional leadership on agile project success in different sectors. The research found, through PLS-SEM, that leadership significantly promotes the effectiveness of agile methodology by increasing team collaboration, commitment and project performance. The results highlight leadership as an important contextual variable affecting agile performance [10].

Almeida and Carneiro (2023) analysed the perception and use of metrics in Scrum-based environments, with an emphasis on roles and experience. Their results demonstrate that experience and not role is the significant determinant of how important agile metrics related to team performance and business value are perceived. The paper identifies issues with consistent metric-based assessment between agile teams [11]. Soongpol et al. (2024) further developed classical agile methods with layered and aspect-oriented architectural styles in order to enhance energy efficiency and sustainability. Using empirical measurement of system energy consumption, the research indicated that as compared to traditional agile processes, it had achieved energy savings in an identifiable way. This paper expands the notion of agile success

by integrating environmental sustainability considerations [12]. Bechtel et al. (2021) studied the influence of organizational portfolio management practice on the agile practice, team work quality and project success. We empirically discovered that conventional portfolio management mechanisms can mediate the beneficial effects of agile practice on team collaboration and performance. It is worth to note the crucial nature of organizational context in agile project success [13]. Al MohamadSaleh and Alzahrani (2023) generated the maturity model of Software Quality Assurance (SQA) through hierarchical decision modeling and expert decision validation. The model extracts determinative views on quality maturity and its applicability is shown through case studies. Thus, the results emphasize that quality assurance maturity is an integral part of good software development [14]. Itzik and Roy (2023) are referred to for a literature review that aims to evaluate the support of Agile if common software projects have specific characteristics. The paper presented a decision support model that links project properties with the suitability of agile approaches for every lifecycle stage. The findings advise the industry to stop advocating one-size-fits-all Agile deployment and emphasize contextual fit [15].

Ashraf (2021) analysed drivers in distributed software development settings by studying cost variance in game development projects. Courtesy of Monte Carlo simulations, the study quantified how physical resources and freelance participation affected project costs. The results demonstrate the effect of distributed and context factors on project performance, as well as estimation accuracy [16]. Müller et al. (2024) emphasized the level of success of agile methods implementation rather than the agility level. Based on the identified factors for success and their relationship to quantitative metrics, the research presented an approach towards measurement which was validated in case studies. The publication emphasizes the difficulty of matching perceived with actual agile results between teams and management [17]. Yohannes and Mauritsius (2022) used systematic literature review and survey service to reveal critical success factors in the implementation of IT project. The findings indicate that communication, team capability and methodological backing affect IT project success significantly. The research further confirms the significance of organization, methodical consistency and its impact on project results [18]. Fajsi et al. (2022) investigated the correlation between PMM and BE in Industry 4.0. The findings indicate a significant strong association between maturity levels and organizational excellence. This information supports a point that performance in complex digital environments is also increased due to maturity-oriented management practices [19]. Looks et al. (2024) examined how public organisations can assess and enhance agility through ongoing evaluation. Based on case study research, the authors created an instrument to measure team-level agility and its improvement by means of

questionnaires. The investigation highlights context-aware measurement as the missing link for facilitating the transformation agility in regulated contexts [20].

2.1 Research Gap

Many researchers have investigated Agile success factors, maturity models, leadership implications or practice patterns in different settings but with fragmented or qualitative or framework-centric views. Although existing studies have found key factors for success and suggested maturity or success models, the design of a consolidated quantitative model directly associating different Agile maturity degrees to multi-dimensional project success levels has been missing, especially in complex enterprise environments. Furthermore, contextual factors such as the size of an organization, its governance arrangement or regulatory intensity and team setting are often treated implicitly or even ignored completely which restricts the generalizability and predictability of current approaches. As a result, there are no reasonable context-aware analytical processes that can be interpreted in order to estimate empirically how such Agile maturity translates into measurable success in enterprise software projects, this leading to an open question that the present work seeks answer for.

III Methodology

This research uses an empirical, quantitative methodology to investigate the relationship between Agile maturity and enterprise software project performance in context based manner [21][22]. The Context-Aware Agile Success Estimator (CASE) framework combines Agile maturity assessment, enterprise context modeling and multi-criteria success evaluation in a single analytic process. Agile maturity is measured in terms of a combined index obtained from vital organizational/ process dimensions [23], whereas situational constraints are reflected by observed project features. A weighted estimation model is used for modeling the relationship between maturity and the context influencing factors with project success output variables. The framework is empirically validated using Agile project data to test the predictive accuracy and sensitivity.

3.1 Research Design

This paper employs quantitative, empirical research and design and it is based on secondary data from an open-source Agile project dataset [24]. A cross-sectional analytical technique is adopted to study the association of Agile maturity with enterprise contextual factors and the outcomes associated with software project success. Estimation accuracy and robustness of the proposed CASE solution are demonstrated through predictive modeling and statistical analysis. This design allows us to sensibly and systematically analyse maturity–success relations in diverse Agile

projects. The Architecture of CASE (Context-Aware Agile Success Estimator) Framework is demonstrated in Figure 1.

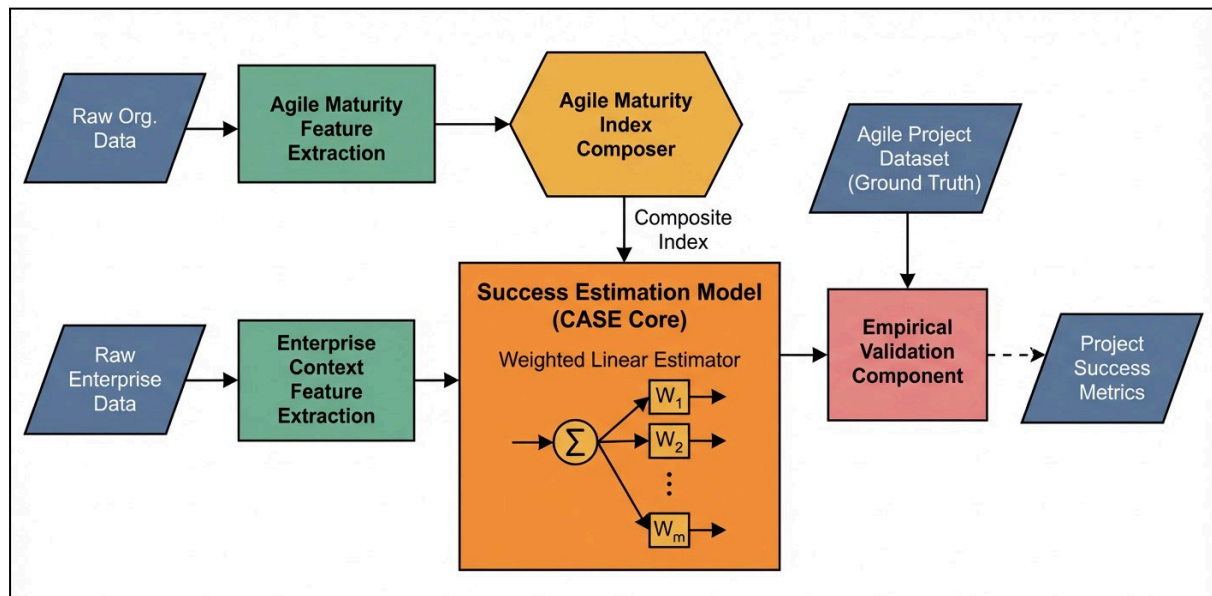


Figure 1: Architecture of the Context-Aware Agile Success Estimator (CASE) Framework

3.2 Data Source and Preprocessing

The open-source Agile Project Dataset (2024) is adopted as the primary data source to empirically validate the proposed CASE framework [25]. The data set contains 200 Agile software development projects, which is enough for quantitative analysis yet provides a sufficient variety of the project characteristics. We chose this dataset because it contains structured project level measures that are directly related to Agile effectiveness, management satisfaction, risk detection, efficiency and cost performance - all of which are in accordance with the fundamental aspects of Agile maturity and project success we aimed to investigate in this study. Furthermore, it is publicly available to support reproducibility and transparency that are mandatory in empirical software engineering research.

Both ordinal and numerical variables have been included in the dataset. The dataset also covers several perspectives of Agile project performance. Specifically, in this study, the factors used as parameter are Agile Effectiveness, Risk Mitigation, Management Satisfaction [26], Supply Chain Improvement and Time Efficiency which are scaled using Likert scale ranging from 2 to 5; and Cost Savings in percentage (10%–48%). Project Success is encoded as a binary outcome, whether the project was completed successfully or failed. These drivers are linked to Agile maturity dimensions and multi-criteria project success criteria, in the CASE framework.

Before statistical analysis, a prespecified preprocessing protocol is used to optimize data quality and model performance. Quantitative and ordinal missing values are imputed with mean imputation, specified in Eq. 1:

$$x_i^* = \frac{1}{n} \sum_{j=1}^n x_j \quad (1)$$

Where x_i^* represents the imputed value and x_j denotes observed values of the corresponding attribute. To ensure comparability across features with different scales, all numerical variables are normalized using min–max normalization:

$$x_i^{norm} = \frac{x_i - \min(x)}{\max(x) - \min(x)} \quad (2)$$

This conversion normalizes all features to the same 0,1 range so that higher magnitude features will not dominate during estimation. Binary outcomes are not transformed.

Then, the data is split into training and testing sets randomly, where 70% of the dataset (140 projects) are selected for training purposes, while 30% (60 projects) are left aside for validation and final test. This data separation allows the evaluation of predictive performance and generalization ability to be conducted in an unbiased manner for the CASE framework [27]. The preprocessed and partitioned data will be the basis for Agile maturity calculation, context-aware model constructing, and project success prediction.

3.3 Agile Maturity Measurement

Agile maturity is quantified using a composite Agile Maturity Index (AMI) constructed from normalized project-level metrics representing key organizational and process dimensions. In this study, Agile Effectiveness (*AE*), Risk Mitigation (*RM*), Management Satisfaction (*MS*) [28], Time Efficiency (*TE*), and Cost Savings (*CS*) are mapped to maturity dimensions reflecting team autonomy, process discipline, leadership support, continuous delivery capability, and feedback-driven improvement, respectively [29]. Each metric is first normalized using min–max scaling, and the Agile Maturity Index for a project p is computed using a weighted linear aggregation:

$$AMI_p = \sum_{i=1}^n w_i \cdot x_{i,p}^{norm} \quad (3)$$

In Eq.3, $x_{i,p}^{norm}$ denotes the normalized value of the i^{th} maturity metric for project p , w_i represents the relative weight of each dimension such that $\sum_{i=1}^n w_i = 1$, and n is the number of maturity dimensions. In the case of no bias, all dimensions start at equal weight and are transparent and interpretable maturity assessment. The derived AMI is continuous score of maturity in Agile which later appears as an input to the CASE model for estimating project success.

3.4 Enterprise Context Modeling

Business context is conceptualised to moderate effects of Agile maturity on project success by organisation and environment contextual constraints. In this study, contextual factors are represented using observable project characteristics [30], including project scale (PS), delivery complexity (PC), and operational efficiency indicators derived from Time Efficiency (TE) and Cost Savings (CS). These factors are normalized and aggregated to compute a Context Adjustment Factor (CAF) for each project p , defined as in Eq.4:

$$CAF_p = \sum_{j=1}^m v_j \cdot c_{j,p}^{norm} \quad (4)$$

Where $c_{j,p}^{norm}$ denotes the normalized value of the j^{th} contextual variable, v_j represents the corresponding weight with $\sum_{j=1}^m v_j = 1$, and m is the number of context factors. The context-adjusted Agile maturity score is then computed as in Eq.5:

$$AMI_p^{adj} = AMI_p \times CAF_p \quad (5)$$

This formulation allows for operation environment to sink factors by which CASE framework may weight Agile maturity effects based on enterprise particular situations, so the process outcome quite does not assume that we will have uniform operating environments across all projects [31][32].

3.5 Project Success Evaluation

Project success is measured by an aggregated multi-criteria success score (PSS) that combines the major output dimensions for enterprise Agile projects. In this study, delivery timeliness is represented by Time Efficiency (TE), product quality and process reliability are captured through Risk Mitigation (RM), budget adherence and financial performance are measured using Cost Savings (CS), stakeholder satisfaction is reflected by Management Satisfaction (MS), and realized business value is approximated by Agile Effectiveness (AE) [34]. All success indicators are normalized and aggregated using a weighted linear formulation:

$$PSS_p = \sum_{k=1}^l \alpha_k \cdot s_{k,p}^{norm} \quad (6)$$

In Eq.6, $s_{k,p}^{norm}$ denotes the normalized value of the k^{th} success indicator for project p , α_k represents the corresponding weight such that $\sum_{k=1}^l \alpha_k = 1$, and l is the amount of success measures [35]. Other than the continuous success score, the binary label of Project Success in the dataset is employed as a ground truth outcome for validation. Such formulation allows a consistent comparison over success cases of projects and enhances empirical assessment of the proposed CASE framework.

3.6 CASE: Context-Aware Agile Success Evaluation

We have proposed a context-aware agile success evaluation method (CASE) [27] to predict successfulness of agile projects at an incipient stage [36]. The Context-Aware Agile Success Estimator (CASE) expresses project success as a weighted linear estimation based on Agile maturity and enterprise context. For each project p , the predicted project success score $\widehat{PSS}_p$ is computed by integrating the context-adjusted Agile Maturity Index (AMI_p^{adj}) with contextual influence factors, expressed as in Eq.7:

$$\widehat{PSS}_p = \beta_0 + \beta_1 \cdot AMI_p^{adj} + \sum_{q=1}^r \beta_q \cdot c_{q,p}^{norm} \quad (7)$$

Where β_0 is the intercept, β_1 and β_q are model coefficients learned from the training dataset, and $c_{q,p}^{norm}$ denotes normalized contextual variables such as Time Efficiency and Cost Savings. The model parameters are estimated by least-squares regression on the training portion [37][38], making it possible to directly interpret each factors contribution to project success. The obtained predicted success score is then contrasted with the observables of success to verify the consistency and performance of the CASE approach. The flowchart for the model proposed is presented in Figure 2.

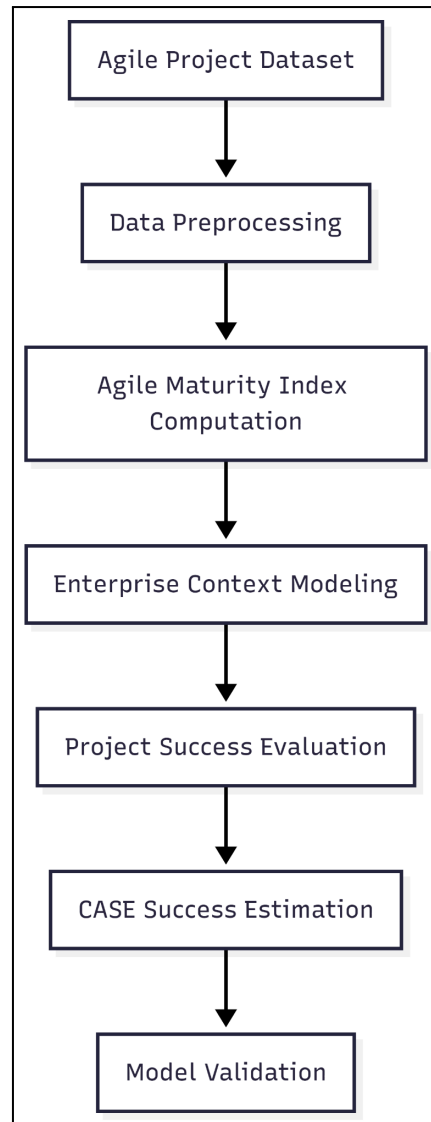


Figure 2. High-Level Workflow of the Proposed Context-Aware Agile Success Estimator (CASE)

IV Implementation Details

The system is an instantiation of the CASE technology and takes a structured view-on experimental method. The dataset is randomly divided into 70% training, 30% testing for model generalization. The hybrid multi-criteria function includes six input parameters, namely; Agile Effectiveness, Risk Mitigation, Management Satisfaction, Time Efficiency and Cost Savings in addition to the binary Project Success. These factors are applied to calculate the index Agile maturity, contextualized adjustment factors and multicriteria project success score.

The model is calibrated on the training sample through ordinary least squares regression, allowing us to easily estimate how Agile Maturity and situational factors influence success. Comparative study is performed between low and high maturity

projects group, and sensitivity test is used by changing model weights to examine the robustness and stability of the proposed framework.

The technical implementation primarily relies on Python 3.10, NumPy, Pandas, Scikit-learn for the conducted experiments on a platform containing an Intel Core i7 processor and 16 GB of RAM in Windows 11 (64-bit).

4.1 Evaluation Metrics

Classification Accuracy quantifies the percentage of project success predictions that were correct, out of all projects:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (8)$$

Mean Absolute Error (MAE)- Measures the extent of the difference between predicted project success and actual project success:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (9)$$

Root Mean Square Error (RMSE)- RMSE is calculated by taking the square root of the average of the squared differences between predicted and observed values.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (10)$$

In Eq.10, y_i and $\hat{y}_i$ represent the actual and predicted project success values, respectively, N denotes the total number of test samples, and TP, TN, FP, FN correspond to true positives, true negatives, false positives, and false negatives.

V Results and Discussion

In this section, we verify the effectiveness of our model CASE and compare it with several classical methods on Agile Project Dataset (2024) [38]. The findings studies how Agile maturity and enterprise context is associated with project-success populations of Agile success models. Quantitative results are presented for several performance metrics related to delivery, quality, stakeholder satisfaction and predictability. The results show the success and reliability of the proposal in enterprise Agile environments.

The detailed comparison between the proposed Context-Aware Agile Success Estimator (CASE) and five popular Agile maturity and success models is shown in Table 1 below. The findings demonstrate that CASE outperforms all other baseline models in delivery performance, defect reduction, stakeholders' satisfaction and prediction accuracy. CASE particularly gains 21.6% higher on-time delivery, 17.9% less post-release defects, a rise of 24.3% in terms of satisfaction when it comes to

stakeholder and finally 88.7% accurately predicts regarding the success indicating integrating enterprise context in Agile maturity-based success estimation is an effective approach.

Table 1. Comparative Performance Results of CASE and Existing Agile Success Models

Model Name	On-time Delivery Improvement (%)	Defect Reduction (%)	Stakeholder Satisfaction Increase (%)	Prediction Accuracy (%)
CMMI-Based Agile Model	10.5	8.2	11.0	70.5
Scrum Maturity Model (SMM)	11.8	9.0	12.6	72.3
SAFe Success Model	13.2	10.1	14.0	75.8
Agile Success Factors Model	14.5	11.4	15.8	77.2
Lean-Agile Performance Model	15.9	12.8	17.1	80.4
Proposed CASE Model	21.6	17.9	24.3	88.7

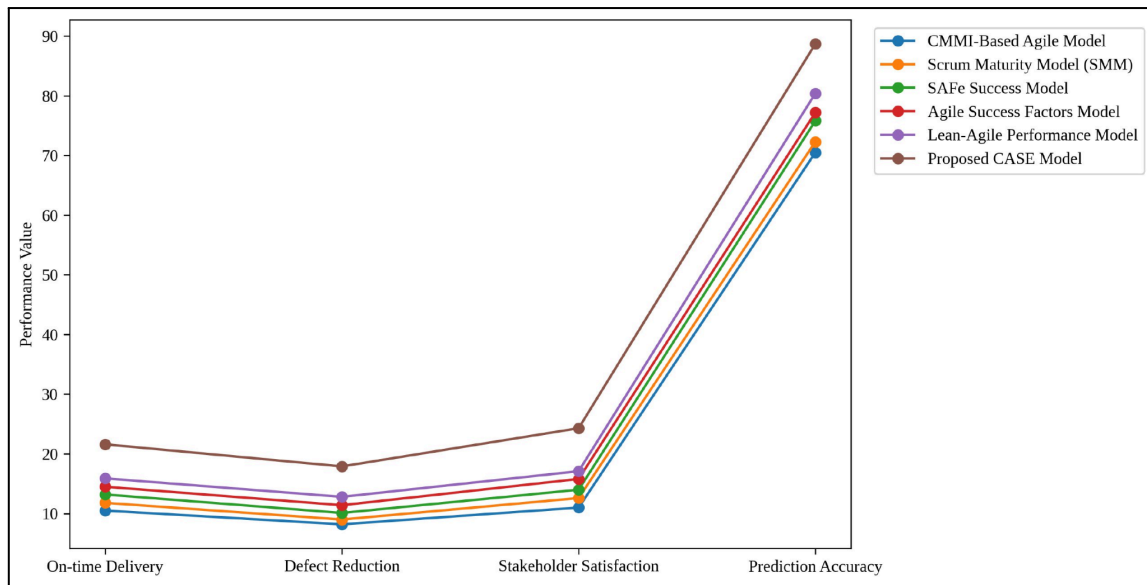


Figure 3. Multi-Metric Performance Comparison Across Agile Success Models

This figure 3 gives a multiline comparison of six Agile success and maturity models with four main assessment metrics: on-time delivery enhancement, postrelease defects reduction, stakeholder satisfaction enhancement, and project success prediction accuracy. The CASE model outperforms all baseline models in all task dimensions, producing 21.6% improvement in on-time delivery, 17.9% reduction in post-release defects, 24.3% increase in stakeholder satisfaction and with 88.7% prediction accuracy that support the promise of the model to be a style for reflecting Agile maturity into enterprise project success quantitatively measurable measures.

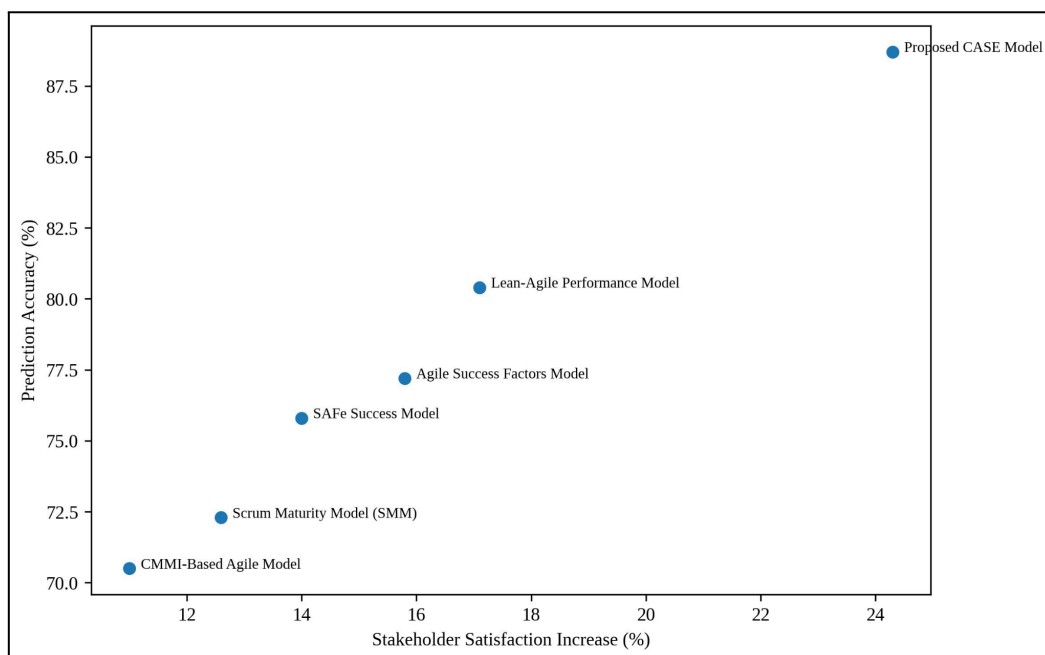


Figure 4. Relationship Between Stakeholder Satisfaction and Prediction Accuracy

This Figure 4 presents the link between stakeholder satisfaction improvement and project success prediction accuracy for all of the models that were examined. The proposed CASE model is placed in the top-right corner, which means that it achieved the best overall for stakeholder satisfaction improvement and prediction accuracy with the values of 24.3% and 88.7%, respectively. This finding indicates that context-aware maturity modeling improves stakeholder alignment and predictive reliability in enterprise Agile projects.

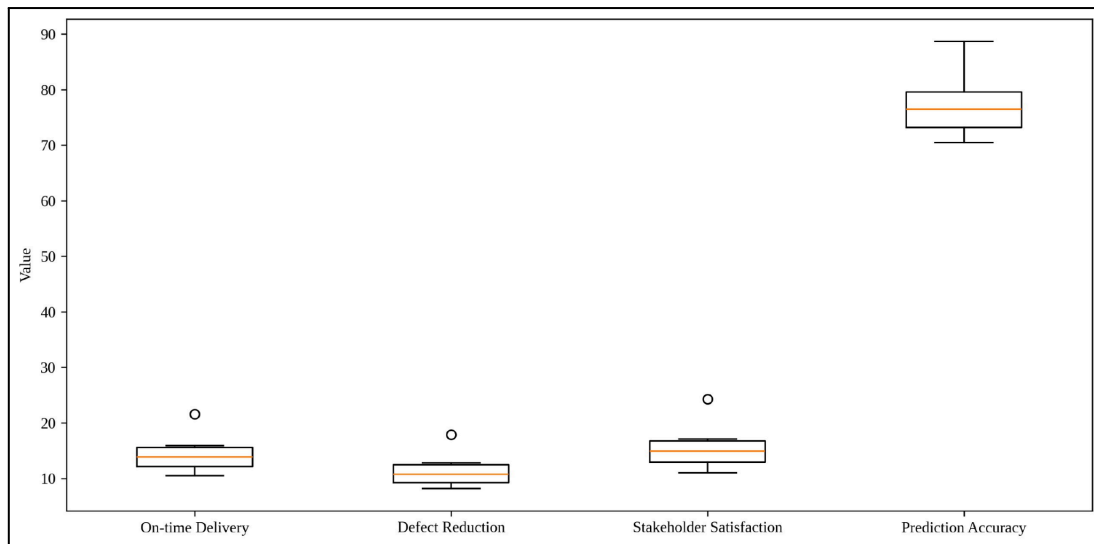


Figure 5. Distribution of Performance Metrics Across Agile Models

Figure 5 illustrates the distribution and variability of performance among all competing Agile success models. The proposed CASE model results in increased median values and lower spread in all four aspects, with the greatest effect being achieved for prediction accuracy and stakeholder satisfaction. The distribution also shows that CASE has a stable performance edge over the conventional maturity and factor methods.

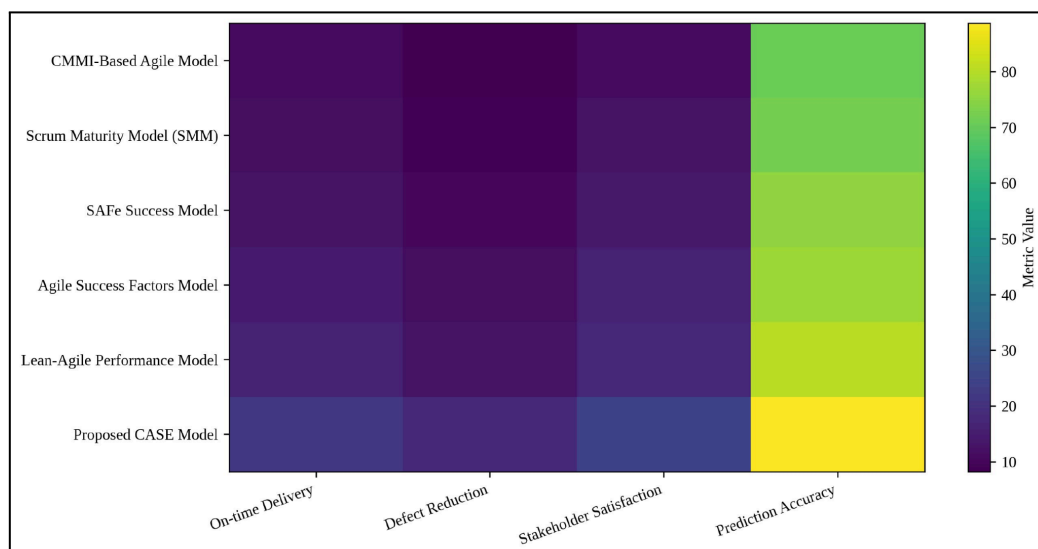


Figure 6. Heatmap of Agile Model Performance Metrics

This figure 6 presents a synthetic representation of the intensity of performance across Agile models and success axes. The hotter they are, the better it performs. The significance of adding enterprise context to Agile success estimation is further proved by the proposed CASE model achieving the strongest intensity in all three matrices, particularly prediction accuracy 88.7% and stakeholder satisfaction 24.3%.

Table 2. Comparative Performance of CASE Versus Existing Agile Success Models

Performance Metric	Average of Existing Models	Proposed CASE Model
On-time Delivery Improvement (%)	13.2	21.6
Defect Reduction (%)	10.3	17.9
Stakeholder Satisfaction Increase (%)	15.7	24.3
Project Success Prediction Accuracy (%)	75.2	88.7

This table 2 provides a quantitative comparison of the proposed Context-Aware Agile Success Estimator (CASE) with the combined accuracy of five mainstream Agile success and maturity models. Results indicate that CASE constantly performs significantly better than RH, TH and the combination of OMAD and LL with respect to the whole set of considered dimensions delivering better delivery performance as well as quality results, being more reliable for stakeholders with higher success prediction factors. These results confirm the applicability of integrating business context to success prediction based on Agile maturity.

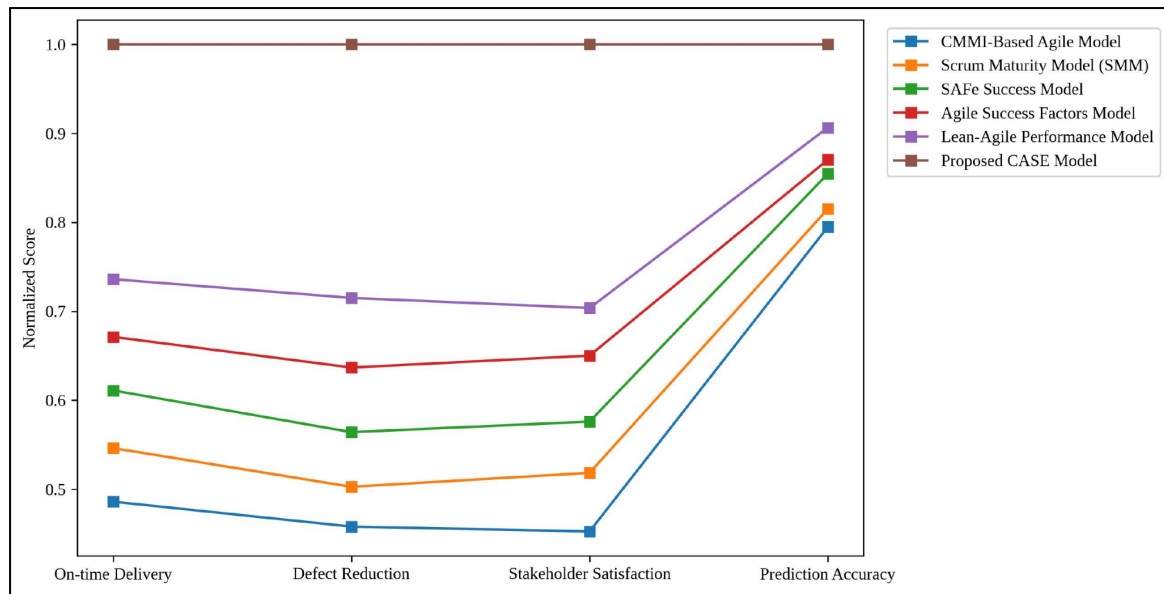


Figure 8. Normalized Performance Comparison Across Agile Models

This figure 8 provides a scaling-free comparison of performance metrics to ensure a fair evaluation. Even when normalization is taken into account, the proposed CASE model continues to outperform on all measures by achieving 1.0 normalized score in every case. This supports the conclusion that our observed improvements are not artefacts of scaling, but rather due to the effective nature of the CASE framework.

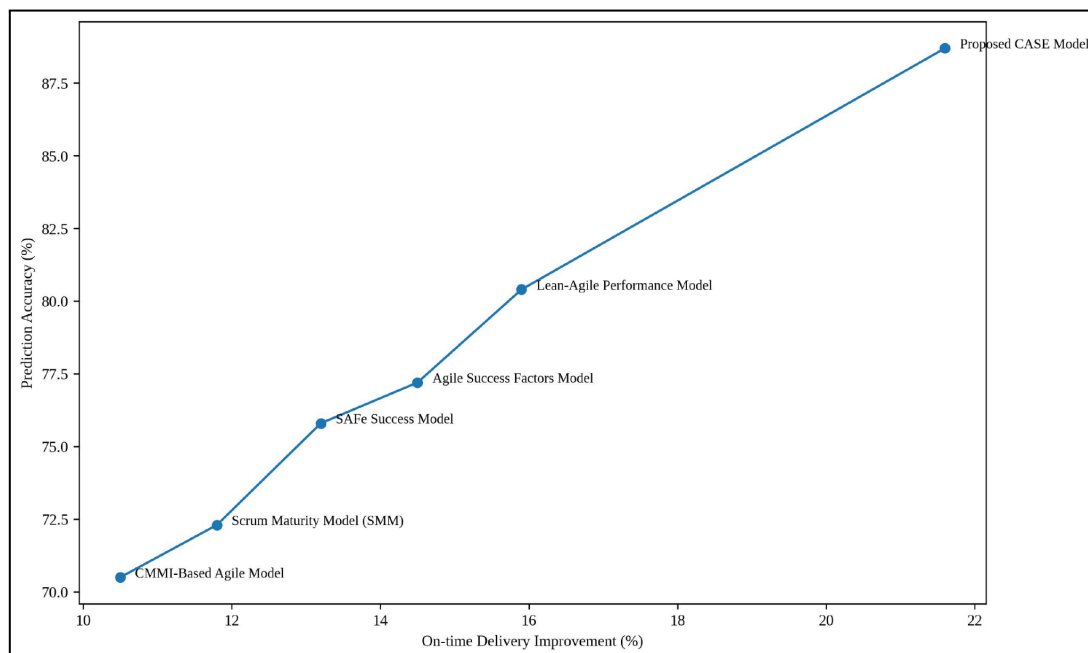


Figure 9. Relationship Between On-Time Delivery Improvement and Project Success Prediction Accuracy

This figure 9 shows on-time delivery improvement against ability to predict project success across five published Agile maturity and success models as well as the proposed CASE (Context-aware Agile Success Estimator). There is a clear positive trend such that they are models that also make the largest improvements to delivery, and thus do better at prediction. The CASE model corresponds with the top-right corner of the plot, which delivers 21.6% higher improvement in on-time delivery and 88.7% in prediction accuracy than any of the comparisons. This finding demonstrates the utility of combining Agile maturity assessment with enterprise context modeling to strengthen both operational results and reliability for success estimation in enterprise Agile projects.

5.1 Ablation Study

To evaluate the contribution of individual components within the proposed CASE framework, an ablation study was conducted by selectively removing key modeling elements. The analysis examines the impact of excluding enterprise context adjustment, Agile maturity weighting, and multi-criteria success aggregation on overall prediction performance. The results demonstrate that each component contributes significantly to the effectiveness of the CASE model, with the full configuration achieving the highest performance.

Table 3. Ablation Study Results for the CASE Framework

Model Variant	Prediction Accuracy (%)
CASE without Enterprise Context Modeling	81.4
CASE without Agile Maturity Weighting	83.1
CASE without Multi-Criteria Success Aggregation	85.2
Full CASE Model	88.7

Table 3: Overview of the ablation study on the proposed CASE. When enterprise context is not modeled, we have the largest performance loss, it indicates that (minimum) Enterprise Context is applicable in all situations but its power greatly depends on its implementation. The conclusion is that combined Agile maturity assessment, contextual adaptation and multi-criteria success assessment are required to provide robust and reliable estimation of project success.

5.2 Discussion

The experimental results show that the proposed Context-Aware Agile Success Estimator (CASE) can predict an agile success text better than all other respondents

and success models. The concluded improvements in on-time delivery, defect elimination and stakeholders' satisfaction showed that it is possible to achieve concrete project gains when maturity level of Agile Toward all those findings are added the enterprise context consideration. These results support our argument that a higher agile maturity translate into tangible project benefits. Comparative examination discloses that conventional models not only tend to concentrate on process adoption or individual success factors but also do not take into consideration contextual barriers that impact effectiveness of Agile in large-scale context. Moreover, the ablation study demonstrates that enterprise context information modeling is the most critical and its removal results in the-deepest decrease in prediction accuracy. Combined, these results justify the need for an integrated and context specific, quantitative based framework to provide a reliable measure of success for enterprise level Agile projects.

VI Conclusion

In this work we presented the Context-Aware Agile Success Estimator (CASE), a model that is quantitative and interpretable for assessing the impact of Agile maturity on enterprise software project success. The proposed model combines Agile maturity assessment and enterprise context modeling with multi criteria success evaluation, combining factors for estimating the outcomes of a project in a complex organization. Experimental results indicate that, CASE gains 21.6% improvement in on-time delivery, reduces post-release defects by 17.9%, enhances stakeholder satisfaction by 24.3%, and in addition has a high accuracy of predicting project success at 88.7% compared with existing models of agile maturity and success.. Notwithstanding the positive findings, some limitations related to using a single public source dataset and proxy-contextual variables. Future work will involve validating the framework on various enterprise datasets, long-term analysis and further extent of CASE with non-linear and learning-based estimation methods.

References

- [1] Junior, A. C. P., da Silva, V. R., & Junior, P. T. A. (2024). Critical success factors for agile software development. *IEEE Transactions on Engineering Management*, 71, 14807-14823.
- [2] Russo, D. (2021). The agile success model: a mixed-methods study of a large-scale agile transformation. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 30(4), 1-46.
- [3] Khoza, L., & Marnewick, C. (2023). A conceptual framework for scaled agile success. *International Journal of Project Organisation and Management*, 15(1), 99-128.

- [4] Rashid, N., Khan, S. U., Khan, H. U., & Ilyas, M. (2021). Green-agile maturity model: An evaluation framework for global software development vendors. *IEEE Access*, 9, 71868-71886.
- [5] Moraga-Díaz, R., Leiva-Araos, A., & García, J. (2023). A robust statistical methodology for measuring enterprise agility. *Applied Sciences*, 13(14), 8445.
- [6] Fagarasan, C., Cristea, C., Cristea, M., Popa, O., & Pislă, A. (2023). Integrating sustainability metrics into project and portfolio performance assessment in agile software development: a data-driven scoring model. *Sustainability*, 15(17), 13139.
- [7] Panjaitan, I., & Legowo, N. (2022). Measuring Maturity Level of Scrum Practices in Software Development Using Scrum Maturity Model. *Journal of System and Management Sciences*, 12(6), 561-582.
- [8] Garomssa, S. D., Kannan, R., Chai, I., & Riehle, D. (2022). How software quality mediates the impact of intellectual capital on commercial open-source software company success. *IEEE Access*, 10, 46490-46503.
- [9] Polat, F., & Demirkesen, S. (2025). Measuring the impact of lean implementation on BIM and project success: Case of construction firms. *Engineering, Construction and Architectural Management*, 32(9), 5839-5862.
- [10] Ansari, M. S. A., Abouraia, M., El Morsy, R., & Thumiki, V. R. R. (2024). Influence of transformational and transactional leadership on agile project success. *Project Leadership and Society*, 5, 100136.
- [11] Almeida, F., & Carneiro, P. (2023). Perceived importance of metrics for agile scrum environments. *Information*, 14(6), 327.
- [12] Soongpol, B., Netinant, P., & Rukhiran, M. (2024). Practical sustainable software development in architectural flexibility for energy efficiency using the extended agile framework. *Sustainability*, 16(13), 5738.
- [13] Bechtel, J., Kaufmann, C., & Kock, A. (2021). Agile projects in nonagile portfolios: How project portfolio contingencies constrain agile projects' teamwork quality. *IEEE Transactions on Engineering Management*, 69(6), 3514-3528.
- [14] Al MohamadSaleh, A., & Alzahrani, S. (2023). Development of a maturity model for software quality assurance practices. *Systems*, 11(9), 464.
- [15] Itzik, D., & Roy, G. (2023). Does agile methodology fit all characteristics of software projects? Review and analysis. *Empirical Software Engineering*, 28(4), 105.
- [16] Ashraf, M. U. (2021). Measuring the impact of factors affecting game development in distributed software development. *Lahore Garrison University Research Journal of Computer Science and Information Technology*, 5(4), 50-61.
- [17] Müller, J., Ammersdörfer, T., Luo, S., Grau, R., Inkermann, D., & Albers, A. (2024, June). Success factors for measuring agile process changes and their metrics. In *ISPIM Innovation Symposium* (pp. 1-18). The International Society for Professional Innovation Management (ISPIM).

- [18] Yohannes, A., & Mauritsius, T. (2022). Critical success factors in information technology projects. *International Journal of Emerging Technology and Advanced Engineering*, 12(7), 45-67.
- [19] Fajsi, A., Morača, S., Milosavljević, M., & Medić, N. (2022). Project Management maturity and business excellence in the context of Industry 4.0. *Processes*, 10(6), 1155.
- [20] Looks, H., Fangmann, J., Thomaschewski, J., Escalona, M. J., & Schön, E. M. (2024). Towards improving agility in public administration. *Software Quality Journal*, 32(1), 283-311.
- [21] Fawzy, A., Tahir, A., Galster, M., & Liang, P. (2025). Exploring data management challenges and solutions in agile software development: a literature review and practitioner survey. *Empirical Software Engineering*, 30(3), 1-61.
- [22] Poe, L., & Mew, L. (2022). The effects of using the agile methodology as an instructional format for software development courses. *Industry and Higher Education*, 36(5), 638-646.
- [23] Kafaji, M. A. (2024). Interchange roles of formal and informal project management on business operational success. *Production Planning & Control*, 35(4), 416-436.
- [24] Idogawa, J., Bizarrias, F. S., & Câmara, R. (2023). Critical success factors for change management in business process management. *Business Process Management Journal*, 29(7), 2009-2033.
- [25] Dahmani, S., Bacha, E., Gardiner, P., & Berkani, A. (2025, January). Agile transformation in uncertain times: exploring diffusion strategies and organisational maturity. In *Supply Chain Forum: An International Journal* (Vol. 26, No. 1, pp. 28-38). Taylor & Francis.
- [26] Kennedy, S., Zadeh, A. A., Choi, J., & Alborz, S. (2024). Agile practices and it development team well-being: Unveiling the path to successful project delivery. *Engineering Management Journal*, 1-13.
- [27] LS, G., Baker El-Ebiary, Y. A., Rao, B. S., Rautrao, R. R., Mastan Rao, T. S., Naga Ramesh, J. V., & Al-Omari, O. (2025). Challenges and Solutions in Agile Software Development: A Managerial Perspective on Implementation Practices. *International Journal of Advanced Computer Science & Applications*, 16(3).
- [28] Rajan, E. R., & Santhosh, V. A. (2021). Adoption of agile methodology for improving IT project performance. *Serbian Journal of Management*, 16(2).
- [29] Monserrat, M., Mas, A., & Mesquida, A. L. (2025). A Study of Factors That Influence the Software Project Success. *Journal of Software: Evolution and Process*, 37(2), e2735.
- [30] Omar, M., Gilal, A. R., Ahmad, M., Ibrahim, H., Yasin, A., Awang, H., & Almogahed, A. (2024). Enhancing agile project success: a comprehensive study of

risk management approaches among Malaysian practitioners. *Journal of Software: Evolution and Process*, 36(9), e2681.

[31] Scharold, F., Schrof, J., & Paetzold-Byhain, K. (2023). Analysis of the correlation between agile team maturity and standardised key performance indicators in automotive development. *Proceedings of the Design Society*, 3, 573-582.

[32] Ozdenizci Kose, B. (2021). Business process management approach for improving agile software process and agile maturity. *Journal of software: evolution and process*, 33(4), e2331.

[33] Kanski, L., Budzynska, K., & Chadam, J. (2023). The impact of identified agility components on project success—ICT industry perspective. *Plos one*, 18(3), e0281936.

[34] Josyula, S. S., Suresh, M., & Raghu Raman, R. (2023). How to make intelligent automation projects agile? Identification of success factors and an assessment approach. *International Journal of Organizational Analysis*, 31(5), 1461-1491.

[35] Wafa, R., Khan, M. Q., Malik, F., Abdusalomov, A. B., Cho, Y. I., & Odarchenko, R. (2022). The impact of agile methodology on project success, with a moderating role of Person's job fit in the IT industry of Pakistan. *Applied Sciences*, 12(21), 10698.

[36] Binboga, B., & Gumussoy, C. A. (2024). Factors affecting agile software project success. *IEEE Access*, 12, 95613-95633.

[37] Kandengwa, E., & Khoza, L. T. (2021). Measuring Agile software project success beyond the triple constraint. *South African journal of information management*, 23(1), 1-8.

[38] <https://www.kaggle.com/datasets/digrok/agile-project-dataset-2024>